

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q14-0304

Septa 138 kV
April 29, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0304. The Interconnection Customer proposes a 105.3 MW generating facility interconnecting at the Septa 138 kV substation in the Pennsylvania area. The request was received on January 28, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 200 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$486,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The BLOUNTSVILLE - TORREY 138 KV is overloaded to 109.9% of its 209 MVA emergency rating for the loss of the RITTER PARK - STEWART MTN DAM 345 KV. The project contributes 6.9% DFAX (7.2 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
TORREY - BLOUNTSVILLE 138 KV	Reconductor line	\$486,000	100% (\$486,000)
	Total allocated to project		\$486,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.